Evaluating Dimension Reduction Approaches for Two-Phase Studies with High-Dimensional Auxiliary Covariates

Osvaldo Espin-Garcia

Department of Epidemiology and Biostatistics, Western University
Dalla Lana School of Public Health and Department of Statistical Sciences, University of Toronto
Department of Biostatistics, University Health Network

International Biometric Conference 2026
Seoul, South Korea
July 14, 2026

Conflict of Interest

I have no current or past relationships with commercial entities.

The Two-Phase Study

Offers a **cost-effective** strategy to collect and analyze expensive covariates by measuring them only in an informative subsample.

Phase I
Whole sample
Collect inexpensive data:
outcome Y, auxiliary Z,
covariate(s) W

→

Phase II
Subsample S_2
(selected using Y, Z)
Collect expensive
covariate(s) X

→

Analysis
Combine Phase I
and Phase II data
(e.g.,SPMLE, AIPW)

The Two-Phase Study

Offers a **cost-effective** strategy to collect and analyze expensive covariates by measuring them only in an informative subsample.

Phase I
Whole sample
Collect inexpensive data:
outcome Y, auxiliary Z,
covariate(s) W

$\rightarrow$

Phase II
Subsample S_2
(selected using Y, Z)
Collect expensive
covariate(s) X

$\rightarrow$

Analysis
Combine Phase I
and Phase II data
(e.g.,SPMLE, AIPW)

Limited work on how to handle high-dimensional auxiliary variables, i.e., Z, in two-phase studies.

Design - Rank-Based Designs

Achieve a phase 2 sample size by selecting an equal number of samples from each of the top and bottom rankings of the following quantities

- Outcome dependent sampling (ODS), i.e. $Y^{(1)}, \ldots, Y^{(N)}$
- Residual dependent sampling (RDS), i.e. $\hat{\epsilon}^{(1)}, \ldots, \hat{\epsilon}^{(N)}$, where $\hat{\epsilon}_i = Y_i - \hat{Y}_i$, $i = 1, \ldots, N$
- Scaled residuals (TZL). Tao, Zeng, and Lin (2020) show that scaling the residuals by $\mathrm{Var}(X|Z)^{1/2}$ (unknown at design stage) render general optimal designs (i.e., minimum variance under the score test).

These designs are appealing because:

1. are unique
2. the selection is intuitive and can be performed quickly
3. for continuous outcomes, no stratification is required

Design - Residual Dependent Sampling

Step 1: Fit a model to predict Y using available covariates in phase one

Step 2: Compute residuals $\hat{\epsilon}_i = Y_i - \hat{Y}_i$

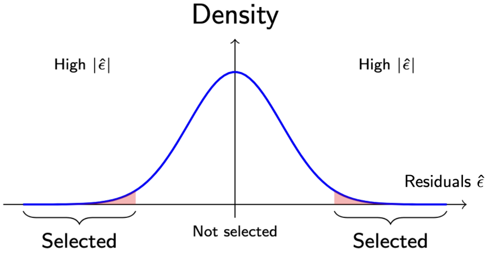
Step 3: Rank subjects by $\hat{\epsilon}_i$

Step 4: Select those with extreme residuals

Intuition:

$\rightarrow$ These are the individuals where model underperformed

$\rightarrow$ Additional data (e.g., X) may improve prediction



Analysis - Likelihood Formulation

Let

- $f_\beta(y|x,z,w)$ be a function member of the exponential family with $\mathrm{E}[Y|X,Y,Z] = h(\beta_0 + \beta_1 X + \beta_2 Z + \beta_3 W)$; $h(\cdot)$ an appropriate inverse link function
- $p_{x,z}$ be the joint probability of $X = x$ and $Z = z$, $\boldsymbol{p} = \{p_{x,z}\}_{x\in\mathcal{X},z\in\mathcal{Z}}$
- $\mathcal{X}$, $\mathcal{Z}$ are the sets of uniquely observed values of X and Z
- $R_i = \mathbb{1}\{i \in S_2\}$ with $\boldsymbol{P}(R_i = 1|y_i, z_i)$ and $\sum_{i=1}^N R_i = n$

The observed data likelihood is defined as

$$L(\beta,\boldsymbol{p}) \propto \prod_{i=1}^{N} \Big[f_\beta(y_i|x_i,z_i,w_i)p(x_i,z_i)\Big]^{R_i}\Big[\sum_{x\in\mathcal{X}} f_\beta(y_i|x,z_i,w_i)p(x,z_i)\Big]^{(1-R_i)}$$

Different estimation approaches:

- Maximum Likelihood (ML) - Espin-Garcia et al, 2018; Zhao et al, 2009; Lawless et al, 1999
- Sieves Maximum Likelihood (SML) - Tao et al, 2017

Analysis - Some Remarks

- The joint probability, $\boldsymbol{p}$, is typically estimated nonparametrically
  - ML requires Z to be discrete to estimate $\boldsymbol{p}$
  - SML uses B-spline basis to approximate $\boldsymbol{p}$ more flexibly, i.e., Z can be continuous
- Maximum likelihood estimates are obtained via the EM algorithm
- The variance-covariance matrix is computed via the Louis' method
- Main interest lies in testing for $\text{H}_0 : \beta_1 = 0$ vs $\text{H}_A : \beta_1 \neq 0$
- In two-phase studies, the expensive covariate, X, is <u>missing by design</u>

Why Two-Phase Designs with High-Dimensional Data?

Challenges:

- Rising cost of collecting expensive covariates
- Existing methods often overlook the complexity inherent to **high-dimensional auxiliary covariates (HACs)**
- May exhibit complex nonlinear relations
- Inferior **subsample selection** and inefficiencies

Objective:

- Evaluate **dimension reduction** methods that transform phase one HACs into univariable predictors that retain essential signals

Dimension Reduction Methods

We reduce the high-dimensional phase-I covariates $\mathbf{Z}$ into a single summary $\tilde{Z} = \hat{h}(\mathbf{Z})$ and use for both RDS (design) and SML (analysis).

We consider four constructions:

- **PCA**: projects $\mathbf{Z}$ onto the direction of maximum variance
- **UMAP**: capture nonlinear structure, manifold-preserving embedding
- **Adaptive LASSO**: sparse linear index learned using Y
- **Meta-Visualization** (Ma et al, 2023): combines multiple embeddings (PCA, MDS, tSNE, etc.) into a stabilized 1-D summary

Simulation Studies - Setup

- $\boldsymbol{Z} \in \mathbb{R}^{N \times q}$: high-dimensional, simulated from a multivariate normal distribution $(q = 20)$
- $W_1 \sim \mathcal{N}(0,1)$, $W_2 \sim$ Bernoulli(0.5), $\boldsymbol{W} = (W_1, W_2)'$
- X (next slide)
- Y is simulated using the following model: $Y = \beta_0 + \beta_1 X + \boldsymbol{W}'\boldsymbol{\beta}_2 + \epsilon$

Parameter	**Settings**
R^2 between $\boldsymbol{Z}$ and X	0.50; 0.75
Phase I sample size	$N = 300$
Phase II sample size	$n = \mathbf{100}; 200$
Effect of X (β_1)	0 (null); 0.35; 0.45
True Transformation – $h(\boldsymbol{Z})$	1) Linear; 2) Weighted Sum; 3) Quadratic; 4) Sine; 5) Combined Nonlinear

Scenario A vs Scenario B: How is X Generated?

Scenario A: Linear Construction from Raw Covariates

- X is generated as a **linear function** of the full high-dimensional covariate vector $\boldsymbol{Z}$, plus Gaussian noise:

$$X_i = \boldsymbol{Z}_i^\top \boldsymbol{\beta}_Z + \varepsilon_i^{(X)}, \quad \varepsilon_i^{(X)} \sim \mathcal{N}(0, \sigma_X^2).$$

- All features in $\boldsymbol{Z}$ contribute to X.
- Noise variance σ_X^2 is chosen to achieve target $R^2 \in \{0.5, 0.75\}$.

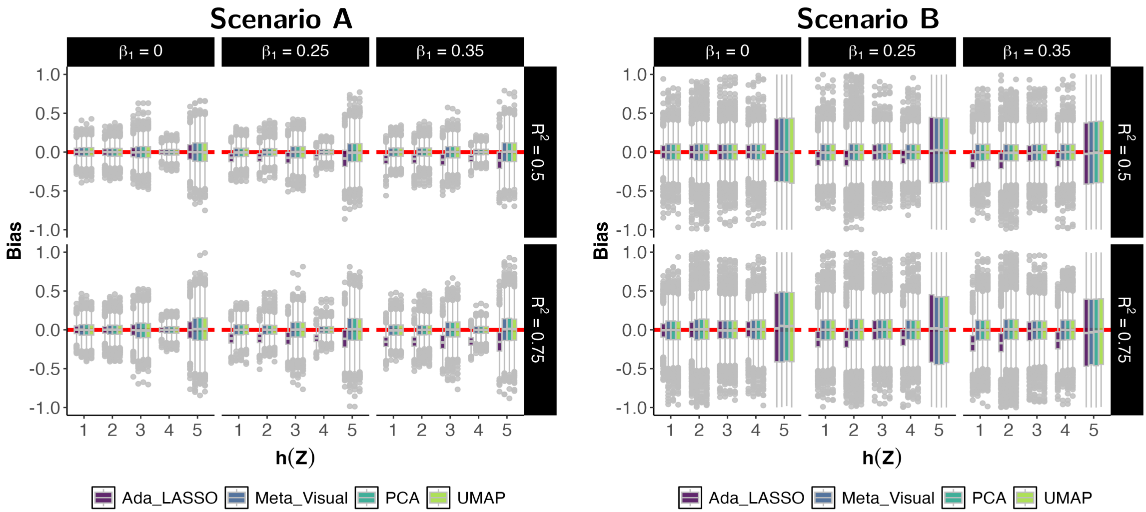
Scenario B: Construction from Transformed Covariates

- X is generated as a linear combination of a **nonlinear transformation** of $\boldsymbol{Z}$, denoted $h(\boldsymbol{Z})$, plus Gaussian noise:

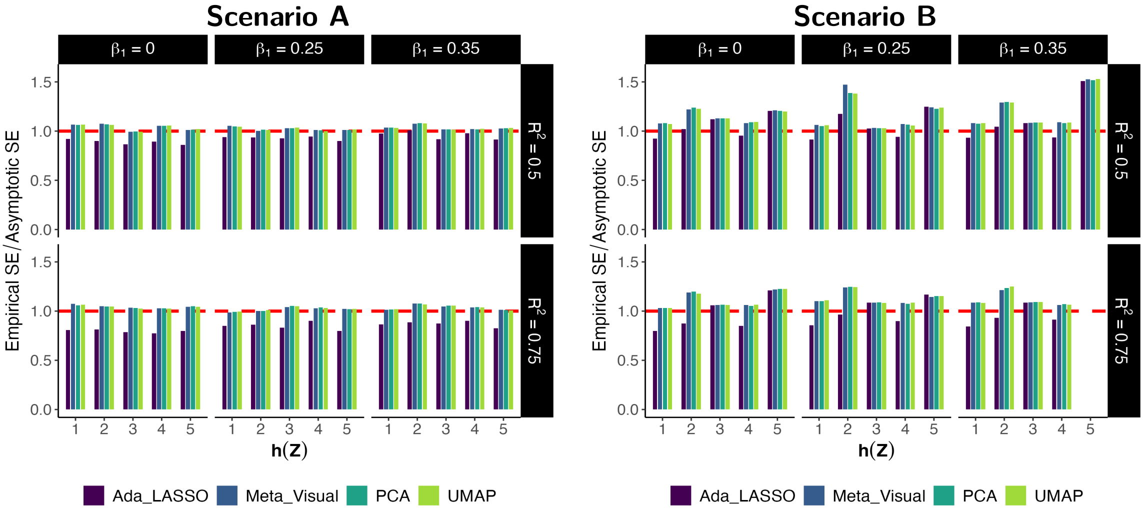
$$X_i = h(\boldsymbol{Z}_i)\beta_{hZ} + \varepsilon_i^{(X)}.$$

- R^2 is controlled with respect to $h(\boldsymbol{Z})$, but the effective correlation between X and the raw covariates $\boldsymbol{Z}$ is not directly constrained.

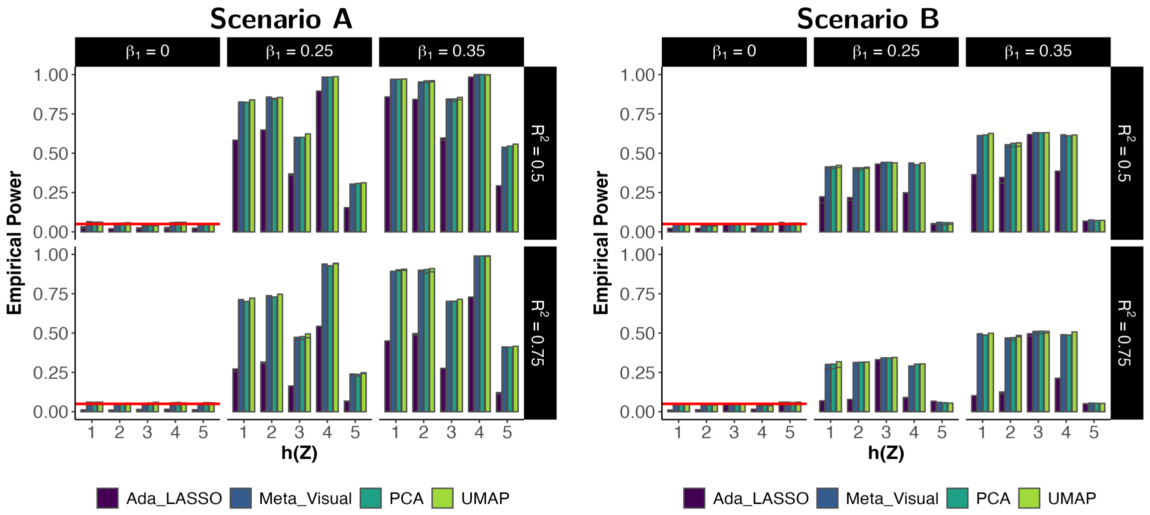
Simulation Results – Bias β_1



Simulation Results – SE Ratio (Empirical / Asymptotic)



Simulation Results – T1E/Power

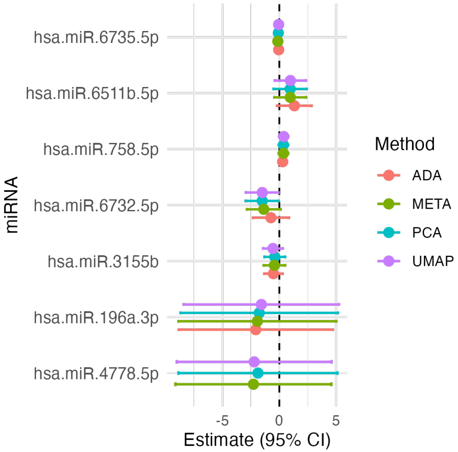


Additional Observations

- **Meta-Visual & UMAP**: best overall (bias, power, SE ratio)
- **PCA performs competitvely across all scenarios**
- **Scenario B Introduce more estimation challeges**

Real Data Application: Osteoarthritis Initiative (OAI)

- **Phase I:** $N = 300$ individuals with baseline data
- **Phase II Sample:** $n = 100$ individuals selected by residual-dependent sampling (RDS)
- **Outcome (Y):** Total Joint Space Width
- **Covariates:**
  - Z: High-dimensional ($q = 25$) MSK predicted gene expression (Phase I)
  - W: 5 clinical covariates (Age, BMI, Sex, Comorbidity Score, WOMAC Total Score)
  - X: 10 selected miRNAs, selected by top R^2 and mean correlation with Z (Phase II only)



Limitations and Future Directions

- **Scalability of B-splines in High Dimensions:**
  B-spline estimation becomes unstable when the dimension of $\boldsymbol{Z}$ increases (e.g., > 5):
  - Exponential growth in number of basis functions
  - Risk of overfitting and numerical instability
  - Increased computational cost in SML
- **Uncertainty from Dimension Reduction:**
  The choice of dimension reduction method and tuning affects $\hat{h}(\boldsymbol{Z})$, introducing additional inferential variability not yet quantified

Limitations and Future Directions

- **Scalability of B-splines in High Dimensions:**
  B-spline estimation becomes unstable when the dimension of $\boldsymbol{Z}$ increases (e.g., > 5):
  - Exponential growth in number of basis functions
  - Risk of overfitting and numerical instability
  - Increased computational cost in SML
- **Uncertainty from Dimension Reduction:**
  The choice of dimension reduction method and tuning affects $\hat{h}(\boldsymbol{Z})$, introducing additional inferential variability not yet quantified
- **Future Directions:**
  - Use supervised methods (e.g., PLS) that incorporate Y in constructing $h(\boldsymbol{Z})$
  - Explore computationally-efficient alternatives to B-splines, e.g., neural networks
  - Account for dimension reduction uncertainty via joint models
  - Extend evaluation to binary, survival, or ordinal outcomes

Acknowledgements

- Hanrui Dou (MSc, Western University)

- This research was possible in part thanks to the support of the Natural Sciences and Engineering Research Council of Canada.

NSERC
CRSNG